A Theory of Two-Dimensional Linear Recurring Arrays

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Abstract—**In this paper, two-dimensional arrays of elements of an arbitrary finite field are examined, especially arrays having maximum-area matrices. We first define two-dimensional linear recurring arrays. In order to study the characteristics of two-dimensional linear recurring arrays, we also define two-dimensional linear cyclic codes. A systematic method of constructing two-dimensional linear recurring arrays having maximum-area matrices is given using the theory of two-dimensional cyclic codes. These arrays, here called $\gamma\beta$-arrays, may be said to be two-dimensional analogs of M-sequences. A $\gamma\beta$-array of area $N_x \times N_y$ exists over $GF(q)$ if and only if N_xN_y is equal to $q^N - 1$ for some positive integer N. Many interesting characteristics of the $\gamma\beta$-array, such as the properties of its autocorrelation function and the properties of the characteristic arrays, are deduced and explained.**

I. Introduction

THE CODES that coding theory has treated in the past have been restricted almost exclusively to codes of one dimension, i.e., codes that can be expressed on only one axis such as the time axis. This restriction seems to be partly due to the fact that currently available devices are not suitable for processing multidimensional information.

Some types of codes that could be described as multidimensional arrays have been studied. Product codes, for example, are codes of this type. However, product codes cannot be called multidimensional codes in the sense of this paper since they have been introduced only in order to construct better one-dimensional codes.

Many useful one-dimensional codes have been constructed to date and are being employed in various applications. This fact suggests that there must be multidimensional codes having distinctive properties that are analogous to those of one-dimensional codes. It is clear that the transmission of multidimensional information does not necessarily require a multidimensional code. However, multidimensional codes can play an important role in some applications. For example, a two-dimensional code conceived as a multidimensional analog of an ordinary M-sequence might be useful for detecting two-dimensional shifts of an array of symbols.

Reed and Stewart [1], Spann [2], and Gordon [3] have made studies of the two-dimensional arrays of a so-called perfect map. The concept of the arrays of a perfect map is almost similar to that of arrays having maximum-area matrices, which are defined in Section II of this paper. Calabro and Wolf [9] have made studies on some two-dimensional arrays having interesting properties and have covered many prior studies of these problems. These studies were quite interesting, but seemed to be somewhat lacking in generality.

We first define two-dimensional linear recurring arrays of elements of an arbitrary finite field. We define also two-dimensional linear cyclic codes and utilize them to analyze the characteristics of two-dimensional linear recurring arrays. A systematic method of constructing two-dimensional arrays having maximum-area matrices is presented using the results obtained through this analysis. These arrays are called $\gamma\beta$-arrays.

These $\gamma\beta$-arrays have many interesting characteristics that can be regarded as two-dimensional analogs of M-sequence properties, but they also have some other interesting characteristics that are not present in M-sequences. Two-dimensional linear recurring arrays and two-dimensional linear cyclic codes include the usual one-dimensional linear recurring sequences and linear cyclic codes.

II. Two-Dimensional Linear Recurring Arrays and Two-Dimensional Linear Cyclic Codes

Two-Dimensional Linear Recurring Arrays

An infinite matrix A_∞ over $GF(q)$ (q is an arbitrary power of an arbitrary prime number)

$$A_\infty = \begin{bmatrix} a_{0,0} & a_{0,1} & a_{0,2} & \cdots \\ a_{1,0} & a_{1,1} & & \\ a_{2,0} & & \ddots & \\ \vdots & & & \end{bmatrix}, \qquad a_{ij} \in GF(q) \quad (1)$$

is called here a two-dimensional array over $GF(q)$. If for arbitrary nonnegative integers I and J, the elements $\{a_{I+l,J+k} \mid k = m \text{ and } 0 \le l < n \text{ or } l = n \text{ and } 0 \le k < m\}$ of A_∞ can be obtained from an $n \times m$ submatrix of A_∞

$$\begin{bmatrix} a_{I,J} & a_{I,J+1} & \cdots & a_{I,J+m-1} \\ \vdots & & & \vdots \\ a_{I+n-1,J} & & \cdots & a_{I+n-1,J+m-1} \end{bmatrix} \quad (2)$$

by applying the linear relations

$$a_{I+l,J+k} = \sum_{i=0}^{n-1} \sum_{j=0}^{m-1} c_{i,j}^{(l,k)} a_{I+i,J+j},$$
$$c_{i,j}^{(l,k)} \in GF(q);\ k = m,\ 0 \le l < n;\ l = n,\ 0 \le k < m \quad (3)$$

Manuscript received July 23, 1971.

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then A_∞ is said to be a two-dimensional linear recurring array. The set of relations (3) is defined to be a two-dimensional linear recurring relation of degree (n,m). The array A_∞ can be generated by this linear recurring relation, if an $n \times m$ matrix that is located at the position (0,0) is given initially.

Equations (3) do not always represent a linear recurring relation if an arbitrary set $\{C_{i,j}^{(l,k)}\}$ of the elements of $GF(q)$ is used as the coefficients of (3), because there are many ways of determining an element $a_{i,j}$ by (3), and these ways must not contradict each other. A necessary and sufficient condition for the set $\{C_{i,j}^{(l,k)}\}$ to define a linear recurring relation without contradiction has already been obtained [4].

It is obvious that all $n + m$ relations of (3) can be obtained from at most $[\min(n,m) + 1]$ relations: $n + 1$ relations determining $\{a_{I+l,J+k} \mid k = m \text{ and } 0 \le l < n \text{ or } k = 0 \text{ and } l = n\}$ for $n \le m$, or $m + 1$ relations determining $\{a_{I+l,J+k} \mid k = m \text{ and } l = 0 \text{ or } 0 \le k < m \text{ and } l = n\}$ for $n \ge m$. These $[\min(n,m) + 1]$ relations may be conveniently used for the purpose of constructing linear recurring arrays.

If there exist positive integers p_x and p_y satisfying

$$a_{i,j} = a_{i+p_x,j} = a_{i,j+p_y} \tag{4}$$

for all $i,j \ge 0$, then the period of A_∞ is denoted by (p_x,p_y), where p_x and p_y are the least positive integers satisfying (4).

An array A_∞ is said to have maximum-area matrices of area $N_x \times N_y$ if, for some positive integers n_x and n_y, all $n_x \times n_y$ submatrices are different from each other and every nonzero $n_x \times n_y$ matrix over $GF(q)$ appears in an arbitrary $(N_x + n_x - 1) \times (N_y + n_y - 1)$ matrix contained in A_∞.

It is obvious that for an $N_x \times N_y$ matrix to be a maximum-area matrix a necessary condition is $N_xN_y = q^{n_xn_y} - 1$.

In this paper, an array having a period (p_x,p_y) and maximum-area matrices of area $N_x \times N_y$ is said to be an M-array if $p_x = N_x$ and $p_y = N_y$.

Two-Dimensional Linear Cyclic Codes

An $N \times M$ matrix over $GF(q)$

$$A = \{a_{i,j}\} = \begin{bmatrix} a_{0,0} & \cdots & a_{0,M-1} \\ \vdots & & \vdots \\ a_{N-1,0} & \cdots & a_{N-1,M-1} \end{bmatrix}, \qquad a_{i,j} \in GF(q) \tag{5}$$

is called an $N \times M$-array. The set $V_{N\times M}$ of all these matrices is an NM-dimensional vector space over $GF(q)$. Every subspace C of $V_{N\times M}$ is called a two-dimensional linear code of area $N \times M$.

The three-dimensional matrix G, which is obtained by piling up $N \times M$ matrices $G_1,G_2,\cdots,G_K$ is called a generator matrix of the linear code C, where K is the dimension of the subspace C and $\{G_i\}$ is a basis of C. Hereafter, G is expressed as $G = \{g_{i,j}\}$ where $g_{i,j}$ are the elements of the K-dimensional vector space. This means that G is regarded as an $N \times M$ matrix over the K-dimensional vector space.

The dual code of C is defined as a two-dimensional linear code that is a null space of C. (The inner product is defined by

$$(A,B) = \sum_{i=0}^{N-1} \sum_{j=0}^{M-1} a_{i,j}b_{i,j},$$

where $A = \{a_{i,j}\}$ and $B = \{b_{i,j}\}$ are two elements of $V_{N\times M}$.) The generator matrix $H = \{h_{i,j}\}$ of this dual code is called a parity-check matrix of C, where the $h_{i,j}$ are elements of $(NM - K)$-dimensional vector space. It is evident that a necessary and sufficient condition for an $N \times M$-array $A = \{a_{i,j}\}$ to be a codeword of C is

$$\sum_{i=0}^{N-1} \sum_{j=0}^{M-1} a_{i,j}h_{i,j} = 0. \tag{6}$$

A two-dimensional linear code C is said to be a two-dimensional linear cyclic code, if and only if any cyclic shift of any codeword $A = \{a_{i,j}\}$ is also a codeword of C. The cyclic shift of $A = \{a_{i,j}\}$ is an $N \times M$-array $\{a_{i+k,j+l}\}$, where k and l are arbitrary integers and the subscripts u and v of $a_{u,v}$ are calculated modulo N and M, respectively.

It is convenient when studying the characteristics of two-dimensional linear cyclic codes to introduce a ring $K[x,y]$, which consists of all two-variable polynomials over $GF(q)$. A residue class ring of $K[x,y]$ with respect to an ideal $(x^N - 1, y^M - 1)$ is denoted by $\Lambda(x^N - 1, y^M - 1)$. There exists a unique polynomial of degree less than N with respect to x and less than M with respect to y in every residue class of $\Lambda(x^N - 1, y^M - 1)$. This can be shown in a way similar to that for the one-dimensional case [4]. A residue class will be represented by such a unique polynomial hereafter. The ring $\Lambda(x^N - 1, y^M - 1)$ is an NM-dimensional vector space over $GF(q)$ and is isomorphic to $V_{N\times M}$ with respect to the correspondence $a(x,y) \leftrightarrow \{a_{i,j}\}$, where $a_{i,j}$ is the coefficient of the term x^iy^j of $a(x,y)$. In this paper, polynomials are always regarded as the elements of $\Lambda(x^N - 1, y^M - 1)$ if not otherwise indicated. Furthermore, a polynomial

$$a(x,y) = \sum_{i=0}^{N-1} \sum_{j=0}^{M-1} a_{i,j}x^iy^j$$

is also considered to represent a two-dimensional array $A = \{a_{i,j}\}$.

A two-dimensional linear code C is a two-dimensional linear cyclic code if and only if $a(x,y) \in C$ implies $x^ky^la(x,y) \in C$, where k and l are arbitrary positive integers. It is easy to prove that a subspace C of the vector space $\Lambda(x^N - 1, y^M - 1)$ is a two-dimensional linear cyclic code if and only if C is an ideal of the residue class ring $\Lambda(x^N - 1, y^M - 1)$ [4].

Now, an infinite two-dimensional array of a finite array A can be regarded as an array A_∞. It is evident that such an array A has a period. Hereafter such an array A_∞ is also, for brevity, denoted by A or $a(x,y)$. The array $A(= A_\infty)$ can always be generated by means of some linear recurring relation if A is a codeword of a two-dimensional linear cyclic code. For example, parity-check polynomials of C [cf. (6)] can be used as the linear recurring relation.

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Examples: Let $q = 2$, $n = 3$, and $m = 2$.

i) The array shown in Fig. 1 is obtained from the linear recurring relation

$$a_{I+3,J} = a_{I,J} + a_{I+2,J}$$
$$a_{I,J+2} = a_{I,J} + a_{I,J+1} + a_{I+2,J+1}. \quad (7)$$

This array is an M-array.

ii) In Fig. 2, the array obtained from the linear recurring relation (8) is shown.

$$a_{I+3,J} = a_{I,J} + a_{I+2,J}$$
$$a_{I,J+2} = a_{I,J+1} + a_{I+1,J} + a_{I+2,J+1}. \quad (8)$$

This array has maximum-area matrices but is not an M-array.

iii) From the linear recurring relation (9) the array that is shown in Fig. 3 is obtained. This array has no maximum-area matrices.

$$a_{I+3,J} = a_{I,J+1} + a_{I+1,J} + a_{I+2,J}$$
$$a_{I+3,J+1} = a_{I,J} + a_{I,J+1}$$
$$a_{I,J+2} = a_{I,J} + a_{I,J+1} + a_{I+1,J+1} + a_{I+2,J+1}. \quad (9)$$

III. $\gamma\beta$-Arrays

Definitions

Let n, m, μ, and λ be positive integers that satisfy the following three conditions:

$$\mu\lambda \mid q^n - 1 \quad (10)$$

$$\text{if } \frac{q^n - 1}{\mu} \Big| q^k - 1 \text{ then } k \geq n \quad (11)$$

$$\gcd\left(\lambda, \frac{q^{nm} - 1}{q^n - 1}\mu\right) = 1, \quad (12)$$

where q is a power of an arbitrary prime number and $a|b$ denotes that b is divisible by a. Then, γ and β are defined by

$$\gamma = \alpha^{[(q^{nm}-1)/(q^n-1)]\mu} \qquad \beta = \alpha^{\eta\lambda}, \quad (13)$$

where α is a primitive element of $GF(q^{nm})$ and η is a positive integer relatively prime to $q^{nm} - 1$. It is obvious that γ is an element of $GF(q^n)$.

It is easy to show that the orders of γ and β are given respectively by

$$e_\gamma = \frac{q^n - 1}{\mu} \qquad e_\beta = \frac{q^{nm} - 1}{\lambda}. \quad (14)$$

Let $h_\gamma(x)$ be a minimal polynomial of γ over $GF(q)$ and $h_\beta(y)$, a minimal polynomial of β over $GF(q^n)$. Then $h_\gamma(x)$ and $h_\beta(y)$, whose degrees are given by Lemma 1, are unique except for multiplication by constants, i.e., multiplication by nonzero elements of $GF(q)$ and $GF(q^n)$, respectively [5].

Lemma 1: The minimal polynomials $h_\gamma(x)$ and $h_\beta(y)$ are irreducible polynomials of degrees n and m, respectively.

Proof: The degree of $h_\gamma(x)$ is given by the least positive integer d satisfying $\mu(q^d - 1) = 0 \bmod (q^n - 1)$ [5]. It follows directly from (11) that $d = n$.

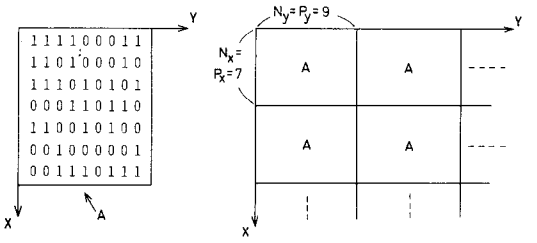


Fig. 1. Example of an M-array (A is a maximum-area matrix).

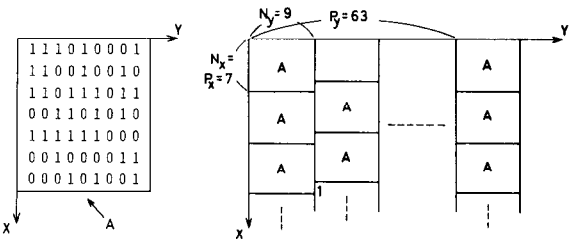


Fig. 2. Example of an array having maximum-area matrices (A is a maximum-area matrix).

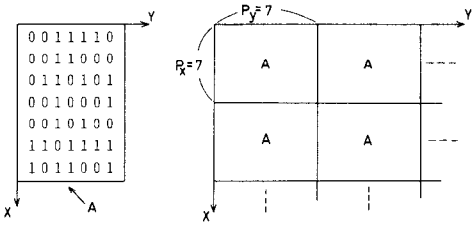


Fig. 3. Example of an array having no maximum-area matrices.

The degree of $h_\beta(y)$ is given by the least positive integer d satisfying $\eta\lambda[(q^n)^d - 1] = 0 \bmod ((q^n)^m - 1)$ [5]. This congruence is equivalent to

$$\lambda(q^{nd} - 1) = 0 \bmod (q^{nm} - 1), \quad (15)$$

because $\gcd(\eta, q^{nm} - 1) = 1$.

Clearly $\lambda < q^n$ since it follows immediately from (10) that $\lambda \mid q^n - 1$. Therefore, $\lambda(q^{nd} - 1) \leq \lambda q^{nd} - 1 < q^{nm} - 1$ if we suppose that $d \leq m - 1$. Then from (15), $\lambda(q^{nd} - 1) = 0$, which is a contradiction. Consequently, d must be greater than or equal to m. Then, the degree of $h_\beta(y)$ is m since $d = m$ satisfies (15). Q.E.D.

Let $P_{n-1} = \{a(x)\}$ be a set of all polynomials over $GF(q)$ of degrees less than n, and let $a(x) \in P_{n-1}$ correspond to $a(\gamma) \in GF(q^n)$. This correspondence between the elements of P_{n-1} and the elements of $GF(q^n)$ is a one-to-one correspondence because $h_\gamma(x)$ is an irreducible polynomial of degree n. Applying this correspondence, let the coefficients $c_i \in GF(q^n)$ of

$$h_\beta(y) = \sum_{i=0}^{m} c_i y^i$$

correspond to $c_i(x) \in P_{n-1}$ and let

$$\sum_{i=0}^{m} c_i(x) y^i$$

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be represented by $h_{\gamma\beta}(x,y)$. Notice here that $h_{\gamma\beta}(y,y) = h_\beta(y)$. Then a set of all elements $f(x,y)$ of $\Lambda(x^{e_\gamma} - 1, y^{e_\beta} - 1)$ satisfying the following simultaneous equations (16) and (17) in $\Lambda(x^{e_\gamma} - 1, y^{e_\beta} - 1)$ is called here a $\gamma\beta$-array code and is represented by $C_{\gamma\beta}$:

$$h_\gamma(x)f(x,y) = 0 \tag{16}$$

$$h_{\gamma\beta}(x,y)f(x,y) = 0. \tag{17}$$

Every nonzero codeword of $C_{\gamma\beta}$ is called a $\gamma\beta$-array.

Structures of a $\gamma\beta$-Array

It is evident from the definition that $C_{\gamma\beta}$ is a two-dimensional linear cyclic code of area $e_\gamma \times e_\beta$. If $f(x,y)$ is expressed as

$$f(x,y) = \sum_{i=0}^{e_\beta - 1} f_i(x)y^i$$

then from (16) we have

$$h_\gamma(x)f_i(x) = 0, \qquad 0 \le i < e_\beta. \tag{18}$$

Consequently the $f_i(x)$ are codewords of a conventional one-dimensional cyclic code C_γ over $GF(q)$ whose parity-check polynomial is $h_\gamma(x)$.

Let $a(x) \in C_\gamma$; then $a(\gamma) \in GF(q^n)$ and the correspondence $a(x) \Leftrightarrow a(\gamma)$ is a one-to-one correspondence. [If $a(x)$, $b(x) \in C_\gamma$, and $a(x) = b(x)$ then $a(\gamma) = b(\gamma)$. Conversely if $a(\gamma) = b(\gamma)$ then $g(x) = a(x) - b(x)$ has all e_γth roots of unity as its roots. Therefore $a(x) = b(x)$.] Accordingly, the correspondence $f(x,y) \Leftrightarrow f(\gamma,y)$ is a one-to-one correspondence, where $f(x,y)$ are the elements of $\Lambda(x^{e_\gamma} - 1, y^{e_\beta} - 1)$ that satisfy (16) and $f(\gamma,y)$ are the one-variable polynomials over $GF(q^n)$.

We can easily show that the correspondence between the solutions $f(x,y)$ of (16), (17) and the solutions $f(\gamma,y)$ of

$$h_\beta(y)f(\gamma,y) = 0 \bmod (y^{e_\beta} - 1) \tag{19}$$

is a one-to-one correspondence, taking notice of the fact that the correspondences $P_{n-1} \Leftrightarrow GF(q^n)$ and $C_\gamma \Leftrightarrow GF(q^n)$ mentioned previously are both isomorphisms. In particular $f(x,y) = 0$ corresponds to $f(\gamma,y) = 0$.

The congruence (19) has q^{nm} solutions, and therefore $C_{\gamma\beta}$ also has q^{nm} codewords, because the degree of $h_\beta(y)$ is m.

Lemma 2: A necessary and sufficient condition for a polynomial $a(x,y)$ to satisfy $a(x,y)f(x,y) = 0$ is $a(\gamma,\beta) = 0$, where $f(x,y)$ is a $\gamma\beta$-array.

Proof: From (19), $f(\gamma,\beta) \neq 0$. (If $f(\gamma,\beta) = 0$ then $f(\gamma,y)$ has all e_βth roots of unity as its roots. Therefore $f(\gamma,y)$ must be zero and then $f(x,y) = 0$.) Consequently, if $a(x,y)f(x,y) = 0$ then $a(\gamma,\beta) = 0$.

Let us regard $h_{\gamma\beta}(x,y)$ as a polynomial in a variable y and let the coefficient of the term y^m be $c(x)$. There exists a polynomial $\bar{c}(x)$, $c(x)\bar{c}(x) = 1 \bmod h_\gamma(x)$ because $c(x)$ is a nonzero polynomial of degree less than n. Then the coefficient of the term y^m of $\bar{c}(x)h_{\gamma\beta}(x,y)$ takes the form $1 + k(x)h_\gamma(x)$. Consequently, any polynomial $a(x,y)$ is divisible by a polynomial $\{\bar{c}(x)h_{\gamma\beta}(x,y) - k(x)h_\gamma(x)y^m\}$ when they are both regarded as polynomials in y. Therefore, $a(x,y)$ can be decomposed as

$$a(x,y) = h_{\gamma\beta}(x,y)a_1(x,y) + h_\gamma(x)a_2(x,y) + a_3(x,y), \tag{20}$$

where the degree of $a_3(x,y)$ with respect to y is less than m. If $a(\gamma,\beta) = 0$ then $a_3(\gamma,\beta) = 0$ from (20). Consequently, $a_3(\gamma,y)$ must be zero. Therefore $a_3(x,y)$ can be decomposed as $a_3(x,y) = h_\gamma(x)a_4(x,y)$. Thus

$$a(x,y) = h_{\gamma\beta}(x,y)a_1(x,y) + h_\gamma(x)\{a_2(x,y) + a_4(x,y)\}.$$

From (16) and (17), $a(x,y)f(x,y) = 0$. Q.E.D.

From this lemma it follows directly that the least positive integers i,j satisfying

$$x^i f(x,y) = f(x,y)$$

$$y^j f(x,y) = f(x,y)$$

are e_γ and e_β, respectively, where $f(x,y)$ is a $\gamma\beta$-array. This means that the period (p_x,p_y) of a $\gamma\beta$-array is

$$(p_x,p_y) = \left(\frac{q^n - 1}{\mu}, \frac{q^{nm} - 1}{\lambda}\right). \tag{21}$$

The following lemma is very important for the purpose of detailed discussion of the structures of $\gamma\beta$-arrays.

Lemma 3: Let $f(x,y)$ be a $\gamma\beta$-array; then

$$x^i y^j f(x,y) = f(x,y), \qquad 0 \le i < p_x, 0 \le j < p_y \tag{22}$$

if and only if

$$i = -\eta\lambda k \bmod p_x,$$

$$j = \frac{q^{nm} - 1}{q^n - 1}\mu k, \qquad 0 \le k < \frac{q^n - 1}{\mu\lambda}. \tag{23}$$

Proof: From Lemma 2, a necessary and sufficient condition for the validity of (22) is

$$\gamma^i\beta^j = 1. \tag{24}$$

Therefore, $\beta^j = \gamma^{-i}$. Taking the e_γth power of both terms, $(\beta^{e_\gamma})^j = 1$. Thus j must be divisible by the order of β^{e_γ}, which is denoted by $e_{\beta\gamma}$. Obviously $e_{\beta\gamma}$ is given by

$$e_{\beta\gamma} = \frac{e_\beta}{\gcd(e_\gamma,e_\beta)} = \frac{q^{nm} - 1}{\lambda \gcd[(q^n - 1)/\mu, (q^{nm} - 1)/\lambda]},$$

while

$$\gcd\left(\frac{q^n - 1}{\mu}, \frac{q^{nm} - 1}{\lambda}\right) = \frac{q^n - 1}{\mu\lambda}\gcd\left(\lambda, \frac{q^{nm} - 1}{q^n - 1}\mu\right)$$

$$= \frac{q^n - 1}{\mu\lambda}$$

because

$$\gcd\left(\lambda, \frac{q^{nm} - 1}{q^n - 1}\mu\right) = 1$$

and

$$\mu\lambda \mid q^n - 1.$$

Therefore,

$$e_{\beta\gamma} = (q^{nm} - 1)\mu/(q^n - 1).$$

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Let $j = ke_{\beta\gamma}$. Then from (13)

$$\gamma^i\beta^j = \gamma^i\beta^{ke_{\beta\gamma}} = \gamma^{i+n\lambda k}.$$

Therefore (24) is valid if and only if

$$i = -\eta\lambda k \bmod e_\gamma,$$
$$j = ke_{\beta\gamma} \bmod e_\beta, \qquad k = 0,1,\cdots. \tag{25}$$

Equation (23) can be derived directly from (25). Q.E.D.

Now, N_x and N_y are defined by

$$N_x = e_\gamma = p_x = \frac{q^n - 1}{\mu} \qquad N_y = e_{\beta\gamma} = \frac{q^{nm} - 1}{q^n - 1}\mu. \tag{26}$$

Corollary 1: Let $f'(x,y)$ be the part of a $\gamma\beta$-array $f(x,y)$ that contains all terms of $f(x,y)$ whose degrees are less than N_y with respect to y. Then $f(x,y)$ can be expressed as

$$f(x,y) = \sum_{k=0}^{\kappa-1} x^{l(k)}y^{N_y k}f'(x,y),$$

where $l(k) = -\eta\lambda k \bmod p_x$ and $\kappa = (q^n - 1)/\mu\lambda$.

From Lemma 3, $x^iy^jf(x,y) = f(x,y)$ if and only if $i = j = 0$ when i and j are restricted to $0 \le i < N_x$ and $0 \le j < N_y$. Furthermore $N_xN_y = q^{nm} - 1$ and there are q^{nm} codewords in $C_{\gamma\beta}$. Thus we have proved the following lemma.

Lemma 4: $C_{\gamma\beta}$ can be expressed using its nonzero codeword $f(x,y)$ as

$$C_{\gamma\beta} = \{x^iy^jf(x,y) \mid 0 \le i < N_x, 0 \le j < N_y\} \cup \{0\}. \tag{27}$$

Now, let us derive the generator matrix of $C_{\gamma\beta}$. Let $\tilde{C}_{\gamma\beta}$ be a set of all $\tilde{a}(x,y)$ such that

$$\tilde{a}(x,y) = x^{p_x}y^{p_y}a(x^{-1},y^{-1}),$$

where $a(x,y)$ is a polynomial that satisfies $a(x,y)f(x,y) = 0$ for a $\gamma\beta$-array $f(x,y)$. It can easily be shown by methods similar to those of one-dimensional coding theory that $\tilde{C}_{\gamma\beta}$ is the dual code of $C_{\gamma\beta}$. From Lemma 2,

$$b(x,y) = \sum_{i=0}^{p_x-1}\sum_{j=0}^{p_y-1} b_{i,j}x^iy^j$$

belongs to $\tilde{C}_{\gamma\beta}$ if and only if $b(\gamma^{-1},\beta^{-1}) = 0$, i.e.,

$$\sum_{i=0}^{p_x-1}\sum_{j=0}^{p_y-1} b_{i,j}\gamma^{-i}\beta^{-j} = 0.$$

Consequently, the parity-check matrix of $\tilde{C}_{\gamma\beta}$, i.e., the generator matrix of $C_{\gamma\beta}$, is expressed as

$$G = \{g_{i,j}\} = \{\gamma^{-i}\beta^{-j}\} = \{\alpha^{-N_yi-\eta\lambda j}\}, \tag{28}$$

where G is a $p_x \times p_y$ matrix over $GF(q^{nm})$.

The principal results thus obtained are summarized in the following theorem.

Theorem 5: The $\gamma\beta$-array code $C_{\gamma\beta}$ over $GF(q)$, which is defined by positive integers n,m,μ,λ satisfying (10), (11), and (12), by a positive integer η that is relatively prime to $q^{nm} - 1$, and by a primitive element α of $GF(q^{nm})$, is a two-dimensional linear cyclic code having q^{nm} codewords. A codeword of $C_{\gamma\beta}$ is a $p_x \times p_y$ array. A generator matrix of $C_{\gamma\beta}$ is given by (28). Finally, we can express $C_{\gamma\beta}$ as in (27), using an arbitrary nonzero codeword $f(x,y)$ of $C_{\gamma\beta}$.

The structure of a $\gamma\beta$-array can be discussed in more detail by investigating the generator matrix of $C_{\gamma\beta}$.

Lemma 6: Let S be a subset having nm elements of the set of all ordered pairs of two integers i and j ($0 \le i < p_x$, $0 \le j < p_y$). Suppose the nm elements $\{g_{i,j} \mid (i,j) \in S\}$ of the generator matrix $G = \{g_{i,j}\}$ of $C_{\gamma\beta}$ are linearly independent over $GF(q)$. Let $f^{(k,l)}(x,y)$ be defined as

$$f^{(k,l)}(x,y) = \sum_{i=0}^{p_x-1}\sum_{j=0}^{p_y-1} f_{i,j}{}^{(k,l)}x^iy^j = x^ky^lf(x,y), \tag{29}$$

where $f(x,y)$ is an arbitrary nonzero codeword of $C_{\gamma\beta}$. Let ω_{kl} be the following ordered sets:

$$\omega_{kl} = \{f_{i,j}{}^{(k,l)} \mid (i,j) \in S\}.$$

Then in the set $\{\omega_{kl} \mid 0 \le k < N_x, 0 \le j < N_y\}$ every ordered set of nm elements of $GF(q)$, except for the set that contains no nonzero element, appears once and only once.

Proof: Let us define the ordered set ω_a for $a(x,y)(\in C_{\gamma\beta})$ by $\omega_a = \{a_{i,j} \mid (i,j) \in S\}$. Then in the set $\{\omega_a \mid a(x,y) \in C_{\gamma\beta}\}$ every ordered set of nm elements of $GF(q)$ appears once and only once, since the dimension of $C_{\gamma\beta}$ is nm and the nm elements $\{g_{ij} \mid (i,j) \in S\}$ of G are linearly independent over $GF(q)$. From this result and Lemma 4, this lemma follows directly. Q.E.D.

The Principal Theorem

The equation

$$\sum_{i=0}^{n-1}\sum_{j=0}^{m-1} \lambda_{i,j}g_{i,j} = \sum_{i=0}^{n-1}\sum_{j=0}^{m-1} \lambda_{i,j}\gamma^{-i}\beta^{-j} = 0, \qquad \lambda_{i,j} \in GF(q)$$

is valid if and only if all $\lambda_{i,j}$ are 0, because the degree of a minimal polynomial of β^{-1} over $GF(q^n)$ is m and the degree of a minimal polynomial of γ^{-1} over $GF(q)$ is n. That is, the nm elements $\{g_{i,j} \mid i = 0, 1,\cdots, n-1$ and $j = 0, 1,\cdots, m-1\}$ of G are linearly independent over $GF(q)$. From this fact and Lemma 6, we can obtain the most important theorem in this paper by setting $n_x = n$ and $n_y = m$ in the definition of arrays having maximum-area matrices.

Theorem 7: A $\gamma\beta$-array has maximum-area matrices of area

$$N_x \times N_y = \left(\frac{q^n - 1}{\mu}\right) \times \left(\frac{q^{nm} - 1}{q^n - 1}\mu\right).$$

In particular, if $\mu\lambda = q^n - 1$, then the $\gamma\beta$-array is an M-array. This array is called a $\gamma\beta$ M-array.

The following corollary is evident from the definition of the maximum-area matrix and Corollary 1.

Corollary 2: In an arbitrary $N_x \times N_y$ array that is included in a $\gamma\beta$-array $f(x,y)$, 0 appears $q^{nm-1} - 1$ times and any other element of $GF(q)$ appears q^{nm-1} times. In a period of $f(x,y)$, 0 appears $(q^{nm-1} - 1)(q^n - 1)/\mu\lambda$ times and any other element of $GF(q)$ appears $q^{nm-1}(q^n - 1)/\mu\lambda$ times.

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The Linear Recurring Relations Generating the $\gamma\beta$-Arrays

The $\gamma\beta$-array can be generated also by a linear recurring relation, which is derived from (16) and (17).

Select minimal polynomials $h_\gamma(x)$ and $h_\beta(y)$ whose constant terms are 1 and γ^{n-1}, respectively. That is, $h_\gamma(0) = 1$ and $h_\beta(0) = \gamma^{n-1}$. Thus, $h_{\gamma\beta}(x,0) = x^{n-1}$. Then $\tilde{h}_i$ is defined as the coefficient of the term x^i of $\tilde{h}_\gamma(x) = x^n h_\gamma(x^{-1})$ and $\tilde{h}_{ij}$ is defined as the coefficient of the term $x^i y^j$ of $\tilde{h}_{\gamma\beta}(x,y) = x^{n-1}y^m h_{\gamma\beta}(x^{-1},y^{-1})$. Equations (16) and (17) can be rewritten as

$$f_{k+n,l} = -\sum_{i=0}^{n-1} \tilde{h}_i f_{k+i,l} \tag{30}$$

$$f_{k,l+m} = -\sum_{i=0}^{n-1}\sum_{j=0}^{m-1} \tilde{h}_{i,j} f_{k+i,l+j}, \tag{31}$$

where $f_{u,v}$ is the coefficient of the term $x^u y^v$ of the $\gamma\beta$-array $f(x,y)$ and the subscripts u and v are calculated modulo p_x and p_y, respectively.

Using (30) and (31), all $f_{i,j}$ can be determined successively if initial values $\{f_{i,j} \mid i = 0, 1, \cdots, n-1$ and $j = 0, 1, \cdots, m-1\}$ are given.

Example: Let $q = 2$, $n = 3$, $m = 2$, $\mu = 1$, $\lambda = q^n - 1 = 7$, $\eta = 1$, and $\alpha^6 + \alpha + 1 = 0$. The $\gamma\beta$-array defined by those parameters is an M-array since $\mu\lambda = q^n - 1$. Further $p_x = N_x = (q^n - 1)/\mu = 7$, $p_y = N_y = (q^{nm} - 1)/\lambda = 9$. The minimal polynomial of $\gamma = \alpha^9$ over $GF(2)$ whose constant term is 1 is $h_\gamma(x) = x^3 + x^2 + 1$ and the minimal polynomial of $\beta = \alpha^7$ over $GF(2^3)$ whose constant term is γ^2 is $h_\beta(y) = \gamma^2(y^2 + (\gamma + 1)y + 1)$. Thus $h_{\gamma\beta}(x,y) = x^2y^2 + y + x^2$. Consequently $\tilde{h}_\gamma(x) = x^3 + x + 1$ and $\tilde{h}_{\gamma\beta}(x,y) = y^2 + x^2y + 1$.

Equations (30) and (31) are expressed as

$$f_{k+3,l} = f_{k,l} + f_{k+1,l}$$

$$f_{k,l+2} = f_{k,l} + f_{k+2,l+1}.$$

From this linear recurring relation the $\gamma\beta$ M-array of Fig. 4 is obtained.

$\gamma\beta$-Arrays Having Maximum-Area Matrices of Area $K \times (q^N - 1)/K$

If a $\gamma\beta$-array over $GF(q)$ is constructed, there must be a positive integer N that satisfies the relation $N_xN_y = q^N - 1$. Furthermore, if the $\gamma\beta$-array is a $\gamma\beta$ M-array, N_x must be relatively prime to N_y, since N_x is equal to λ in that case.

Now, as a converse of the results so far obtained the next theorem can be derived.

Theorem 8: A $\gamma\beta$-array having maximum-area matrices of area $K \times L$ exists if K is an arbitrary divisor of $q^N - 1$ and $L = (q^N - 1)/K$ for some positive integer N. Furthermore a $\gamma\beta$ M-array having maximum-area matrices of area $K \times L$ exists if in addition to the previously mentioned conditions K is relatively prime to L.

Proof: Let n be the least positive integer satisfying the conditions $K \mid q^n - 1$ and $n \mid N$. It is obvious that such an integer n always exists. Then let μ be defined by $\mu = (q^n - 1)/K$ and let λ be a divisor of K where gcd $(\lambda, (q^N - 1)/K) = 1$. At least one such λ exists. For example, let λ be equal to 1. Then it can be shown easily that these n, m, μ, and λ satisfy (10), (11), and (12).

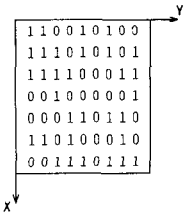


Fig. 4. Example of a $\gamma\beta$-M-array ($q = 2$, $n = 3$, $m = 2$, $\mu = 1$, $\lambda = 7$, $\eta = 1$, $\alpha^6 + \alpha + 1 = 0$).

Let η be an arbitrary positive integer relatively prime to $q^N - 1$ and let α be an arbitrary primitive element of $GF(q^N)$; then the $\gamma\beta$-array constructed using these parameters has maximum-area matrices of area $K \times L$.

If K is relatively prime to L, then λ can be equated to K. Then, $\mu\lambda = q^n - 1$, and in this case the $\gamma\beta$-array previously described is a $\gamma\beta$ M-array.

A Correspondence Between $\gamma\beta$-Arrays and M-Sequences

The minimal polynomial of α over $GF(q)$ is denoted by $h_\alpha(z)$. The cyclic code C_α of length $q^{nm} - 1$ that is defined by the parity-check polynomial $h_\alpha(z)$ is called the M-sequence code [5] or the shortened first-order Reed–Muller code [6]. The following theorem indicates a correspondence between the cyclic code C_α and the $\gamma\beta$-array code $C_{\gamma\beta}$.

Theorem 9: Let $P_{e_\alpha - 1}$ be the set of all polynomials $a(z)$ over $GF(q)$ whose degrees are less than e_α, where $e_\alpha = q^{nm} - 1$, and let P be the set of all polynomials $a'(x,y)$ over $GF(q)$ whose degrees are less than N_x with respect to x and less than N_y with respect to y. Furthermore, let Q be the set of all polynomials $a(x,y)$ defined by the following equation:

$$a(x,y) = \sum_{k=0}^{\kappa-1} x^{l(k)} y^{N_y k} a'(x,y),$$

$$l(k) = -\eta\lambda k \bmod p_x, \quad \kappa = (q^n - 1)/\mu\lambda.$$

Now, let θ be a mapping from Q in $P_{e_\alpha - 1}$ defined by

$$\theta\colon a(x,y) \to a(z) = a'(z^{N_y}, z^{\eta\lambda}) \bmod (z^{e_\alpha} - 1),$$

where $a'(x,y) \in P$ is the part of $a(x,y)$ whose degree with respect to y is less than N_y.

Then θ is a one-to-one mapping from Q onto $P_{e_\alpha - 1}$. The $\gamma\beta$-array code $C_{\gamma\beta}$ is contained in Q and the image of $C_{\gamma\beta}$ is an M-sequence code C_α. Conversely, the inverse image of any M-sequence code of length e_α that is included in $P_{e_\alpha - 1}$ is a $\gamma\beta$-array code.

Proof: It is obvious that θ is a composition mapping $\theta_2 \circ \theta_1$, where

$$\theta_1\colon a(x,y) \to a'(x,y)$$

$$\theta_2\colon a'(x,y) \to a(z) = a'(z^{N_y}, z^{\eta\lambda}) \bmod (z^{e_\alpha} - 1).$$

Obviously θ_1 is a one-to-one mapping onto P. On the other

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hand, θ_2 transfers the coefficient of the term $x^i y^j$ $(0 \le i < N_x, 0 \le j < N_y)$ of $a'(x,y)$ to the coefficient of the term z^l $(l = N_y i + \eta\lambda j \bmod e_\alpha)$ of $a(z)$. Now let

$$l_1 = N_y i_1 + \eta\lambda j_1 \bmod e_\alpha, \qquad 0 \le i_1, i_2 < N_x;$$
$$l_2 = N_y i_2 + \eta\lambda j_2 \bmod e_\alpha, \qquad 0 \le j_1, j_2 < N_y.$$

If $l_1 = l_2$, $\alpha^{l_1} = \alpha^{l_2}$. Thus $\gamma^{i_1}\beta^{j_1} = \gamma^{i_2}\beta^{j_2}$, i.e., $\gamma^{i_1 - i_2}\beta^{j_1 - j_2} = 1$. Meanwhile, from the Proof of Lemma 3 (25) this is valid if and only if $i_1 = i_2$ and $j_1 = j_2$ when i_1, j_1, i_2, and j_2 are restricted by $-N_x < i_1 - i_2 < N_x$, $-N_y < j_1 - j_2 < N_y$. Furthermore $N_x N_y = q^{nm} - 1 = e_\alpha$. Consequently θ_2 is a one-to-one map from (i,j) $(0 \le i < N_x, 0 \le j < N_y)$ to l $(0 \le l < e_\alpha)$. This means that the coefficients of $a(z)$ are obtained by the rearrangement of the coefficients of $a'(x,y)$. Thus it is shown that $\theta = \theta_2 \circ \theta_1$ is a one-to one mapping from Q onto $P_{e_\alpha - 1}$.

From Corollary 1 the codeword $f(x,y)$ of $C_{\gamma\beta}$ is contained in Q. The element of the generator matrix G of $C_{\gamma\beta}$ that corresponds to the coefficient $f_{i,j}$ of $f(x,y)$ is $\alpha^{-N_y i - \eta\lambda j} = \alpha^{-l}$ and the image of $C_{\gamma\beta}$ has the following generator matrix:

$$[\alpha^0 \alpha^{-1} \alpha^{-2} \cdots \alpha^{-e_\alpha + 1}],$$

where α^{-k} is regarded as an nm-dimensional column vector over $GF(q)$. This matrix is the generator matrix of C_α itself. It can easily be shown that $P_{e_\alpha - 1}$ contains no M-sequence code of length e_α except for the images of $\gamma\beta$-array code. From this, the converse is apparent. Q.E.D.

Thus, we can construct $\gamma\beta$-arrays from M-sequences of length $q^{nm} - 1$ over $GF(q)$ by applying the inverse mapping of θ.

IV. The Autocorrelation Functions of $\gamma\beta$-Arrays

The autocorrelation function of the $\gamma\beta$-array has many important properties from the viewpoint of practical applications. First, the autocorrelation function of two-dimensional arrays is defined. Let ϕ be a mapping from the direct product of $GF(q)$ into the set of all real numbers:

$$\phi: (a,b) \mapsto \phi(a,b),$$

where ϕ is assumed to satisfy the three conditions:

$$\phi(a,b) = \phi(b,a) \tag{32}$$
$$\phi(a,a) > 0 \tag{33}$$
$$\sum_{a \in GF(q)} \sum_{b \in GF(q)} \phi(a,b) = 0. \tag{34}$$

The actual form of the function ϕ will be determined by the physical meaning of the elements of $GF(q)$. In most cases, however, the previously mentioned restrictions will be adequate.

The autocorrelation function $\rho_\phi(\tau_1,\tau_2)$ of an array $a(x,y)$ is defined by

$$\rho_\phi(\tau_1,\tau_2) = \sum_{i=0}^{p_x - 1} \sum_{j=0}^{p_y - 1} \phi(a_{i,j}, a_{i+\tau_1, j+\tau_2})/s, \tag{35}$$

where (p_x,p_y) is the period of the $\gamma\beta$-array and the subscripts u,v of $a_{u,v}$ are calculated modulo p_x and p_y, respectively. Here s is given by

$$s = \sum_{i=0}^{p_x - 1} \sum_{j=0}^{p_y - 1} \phi(a_{i,j}, a_{i,j}).$$

From (33), s is always positive. Let us define $\chi(a)$ for $a \in GF(q)$ by

$$\chi(a) = \sum_{b \in GF(q)} \phi(b, ab), \qquad a \in GF(q). \tag{36}$$

Further let

$$\sigma = q^{nm-1}\chi(1) - \phi(0,0). \tag{37}$$

Then, the autocorrelation functions of $\gamma\beta$-arrays can be given by the following theorem.

Theorem 10: The autocorrelation function $\rho_\phi(\tau_1,\tau_2)$ of a $\gamma\beta$-array in a period (i.e., for $0 \le \tau_1 < p_x$, $0 \le \tau_2 < p_y$) is given as follows.

If

$$\tau_1 = -\eta\lambda k + \kappa_1 e l \bmod p_x,$$
$$\tau_2 = N_y k + \kappa_2 e l \bmod p_y, \qquad 0 \le k < (q^n - 1)/\mu\lambda, \quad 0 \le l < q - 1 \tag{38}$$

then

$$\rho_\phi(\tau_1,\tau_2) = \{q^{nm-1}\chi(\delta^l) - \phi(0,0)\}/\sigma,$$

where κ_1 and κ_2 are the integers satisfying

$$N_y \kappa_1 + \eta\lambda\kappa_2 = 1 \bmod (q^{nm} - 1) \tag{39}$$

e is given by $e = (q^{nm} - 1)/(q - 1)$ and δ is a primitive element of $GF(q)$ that is defined by $\delta = \alpha^e$. For values of τ_1 and τ_2 other than those of (38),

$$\rho_\phi(\tau_1,\tau_2) = -\phi(0,0)/\sigma. \tag{40}$$

Remark: From (12), $\gcd(N_y, \eta\lambda) = 1$. Thus integers κ_1 and κ_2 satisfying (39) can easily be obtained using Euclid's algorithm.

Proof: From Corollary 2, it is easily shown that

$$s = \{q^{nm-1}\chi(1) - \phi(0,0)\}(q^n - 1)/\mu\lambda. \tag{41}$$

Then, $\rho_\phi(\tau_1,\tau_2)$ of (35) is rewritten as

$$\rho_\phi(\tau_1,\tau_2) = \sum_{i=0}^{p_x - 1} \sum_{j=0}^{p_y - 1} \phi(f_{0,0}^{(i,j)}, f_{\tau_1,\tau_2}^{(i,j)})/s,$$

where $f^{(i,j)}(x,y)$ was defined previously in (29) and $f_{k,l}^{(i,j)}$ is the coefficient of the term $x^k y^l$ of $f^{(i,j)}(x,y)$. From Corollary 1, (37), and (41) we have

$$\rho_\phi(\tau_1,\tau_2) = \sum_{i=0}^{N_x - 1} \sum_{j=0}^{N_y - 1} \phi(f_{0,0}^{(i,j)}, f_{\tau_1,\tau_2}^{(i,j)})/\sigma. \tag{42}$$

Suppose $\gamma^{\tau_1}\beta^{\tau_2} \in GF(q)$; then from Lemma 2

$$x^{\tau_1} y^{\tau_2} f(x,y) = \gamma^{\tau_1}\beta^{\tau_2} f(x,y).$$

Thus

$$\rho_\phi(\tau_1,\tau_2) = \sum_{i=0}^{N_x - 1} \sum_{j=0}^{N_y - 1} \phi(f_{0,0}^{(i,j)}, \gamma^{\tau_1}\beta^{\tau_2} f_{0,0}^{(i,j)})/\sigma$$
$$= \sum_{i=0}^{N_x - 1} \sum_{j=0}^{N_y - 1} \phi(f_{i,j}^{(0,0)}, \gamma^{\tau_1}\beta^{\tau_2} f_{i,j}^{(0,0)})/\sigma.$$

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From Corollary 2 and (36), it can easily be shown that $\rho_\phi(\tau_1,\tau_2)$ becomes

$$\rho_\phi(\tau_1,\tau_2) = \{q^{nm-1}\chi(\gamma^{\tau_1}\beta^{\tau_2}) - \phi(0,0)\}/\sigma.$$

Now, let us obtain the τ_1 and τ_2 that satisfy $\gamma^{\tau_1}\beta^{\tau_2} \in GF(q)$. If

$$\gamma^{\tau_1}\beta^{\tau_2} = \delta^l = \alpha^{el}, \qquad 0 \le l < q-1$$

then

$$N_y\tau_1 + \eta\lambda\tau_2 = el \bmod (q^{nm}-1). \tag{43}$$

Multiplying both sides by κ_1 and using (39) we have

$$\tau_1 - \eta\lambda\kappa_2\tau_1 + \kappa_1\eta\lambda\tau_2 = \kappa_1 el \bmod (q^{nm}-1).$$

From this equation

$$\tau_1 = \kappa_1 el \bmod \lambda$$

because $\lambda \mid q^{nm}-1$. Similarly

$$\tau_2 = \kappa_2 el \bmod N_y.$$

Then let $\tau_1 = \kappa_1 el + \lambda k_1$ and $\tau_2 = \kappa_2 el + N_y k_2$. Thus from (43)

$$k_1 = -\eta k_2 \bmod (q^n-1)/\mu\lambda.$$

Consequently, setting $k_2 = k$, we can obtain (38) immediately. For τ_1 and τ_2 other than (38), $\gamma^{\tau_1}\beta^{\tau_2} \notin GF(q)$ and, in this case, it is obvious that we can choose nm elements of the generator matrix G of $C_{\gamma\beta}$ including $g_{0,0}$ ($\in GF(q)$) and g_{τ_1,τ_2} ($= \gamma^{-\tau_1}\beta^{-\tau_2} \notin GF(q)$) that are linearly independent over $GF(q)$. Consequently, from Lemma 6, in the set $\{(f_{0,0}^{(i,j)}, f_{\tau_1,\tau_2}^{(i,j)}) \mid 0 \le i < N_x,\ 0 \le j < N_y\}$ (0,0) appears $q^{nm-2}-1$ times and every other combination of two elements of $GF(q)$ appears q^{nm-2} times. Now from (34) and (42), (40) is easily derived. Q.E.D.

This theorem indicates the following properties of the autocorrelation function of the $\gamma\beta$-array.

i) $\rho_\phi(\tau_1,\tau_2)$ assumes at most q values.

ii) In the region $0 \le \tau_1 < N_x$ and $0 \le \tau_2 < N_y$, the number of points where the value of $\rho_\phi(\tau_1,\tau_2)$ is not equal to $-\phi(0,0)/\sigma$ does not exceed $q-1$.

iii) Let (τ_1^0,τ_2^0) be an arbitrary element of the set T^2 that is defined as the set of all (τ_1,τ_2) satisfying (38). Then the set $\{(\tau_1-\tau_1^0, \tau_2-\tau_2^0) \mid (\tau_1,\tau_2) \in T^2\}$ is equal to T^2, if (τ_1,τ_2) is calculated modulo (p_x,p_y). In this sense, the points that have values of $\rho_\phi(\tau_1,\tau_2)$ not equal to $-\phi(0,0)/\sigma$ are uniformly distributed in the $\gamma\beta$-array.

Example: Let $q = 2$, $\phi(0,0) = \phi(1,1) = 1$, and $\phi(1,0) = \phi(0,1) = -1$, then $\chi(1) = 2$, $e = 0 \bmod (2^{nm}-1)$, $\delta = 1$, and $\sigma = 2^{nm}-1$. Consequently,

$$\rho_\phi(\tau_1,\tau_2) = \begin{cases} 1; & \tau_1 = -\eta\lambda k \bmod p_x,\ \tau_2 = N_y k,\ 0 \le k < (q^n-1)/\mu\lambda \\ -1/(2^{nm}-1); & \text{in all other cases.} \end{cases} \tag{44}$$

V. Some Other Properties of $\gamma\beta$-Arrays

The $\gamma\beta$-arrays have many other interesting properties. In this section some of them are stated without proof. It is not difficult to prove these properties, if one recalls the method of constructing $\gamma\beta$-arrays. These properties are closely analogous to those of one-dimensional M-sequences [7].

A Relation Between $\gamma\beta$-Array Codes and Cyclic Product Codes

A direct product code is a kind of two-dimensionally arrayed code. The following theorem makes clear the relation between a $\gamma\beta$-array code $C_{\gamma\beta}$ and a cyclic product code [8].

Theorem 11: Let the minimal polynomials of γ and β over $GF(q)$ be denoted by $h_\gamma(x)$ and $h_\beta'(y)$, respectively. Let cyclic codes C_γ and C_β' of lengths p_x and p_y be defined by parity-check polynomials $h_\gamma(x)$ and $h_\beta'(y)$, respectively. Then the $\gamma\beta$-array code $C_{\gamma\beta}$ is a subcode of the cyclic product code $C_\gamma \times C_\beta'$. Further, if the relation $C_{\gamma\beta} = C_\gamma \times C_\beta'$ holds, p_x must be a divisor of $q-1$.

The Macroscopic Parameters and the Symmetric Images of a $\gamma\beta$-Array

It has been shown that an arbitrary $p_x \times p_y$ subarray of a $\gamma\beta$-array is constructed of $p_y/N_y = (q^n-1)/\mu\lambda$ maximum-area matrices (cf. Lemma 3). The parameters p_x,p_y, N_x,N_y and the mutual positions of p_y/N_y maximum-area matrices (i.e., the magnitude of the cyclic shift along the x axis, $-\eta\lambda \bmod p_x$, of two adjacent matrices) are called the macroscopic parameters of the $\gamma\beta$-array (or of the $\gamma\beta$-array code). An example of the macroscopic parameters of a $\gamma\beta$-array is shown in Fig. 5.

If among the parameters n, m, μ, λ, η, and α that determine the $\gamma\beta$-array code, n, m, μ, and λ are given, then p_x, p_y, N_x, and N_y are determined uniquely [cf. (21) and (26)]. Moreover, if η is given in addition, the mutual positions of the p_y/N_y maximum-area matrices are determined uniquely (cf. Lemma 3). The primitive element α has no effect on those macroscopic parameters. If n,m are equal and μ and/or λ are different in two $\gamma\beta$-array codes, their macroscopic parameters cannot be the same. The macroscopic parameters are not necessarily different, however, when only η is different. Hereafter the $\gamma\beta$-array code having the parameters n, m, μ, λ, η, and α is denoted by C: $\langle n,m,\mu,\lambda,\eta,\alpha\rangle$.

The following theorem shows the relation between two $\gamma\beta$-array codes whose macroscopic parameters are the same.

Theorem 12: A necessary and sufficient condition that a $\gamma\beta$-array code C': $\langle n,m,\mu,\lambda,\eta,\alpha^\nu\rangle$ coincides with the $\gamma\beta$-array code C: $\langle n,m,\mu,\lambda,\eta,\alpha\rangle$ is that $\eta = \eta' \bmod (q^n-1)/\mu\lambda$ and ν is given by

$$\nu = q^k[(1 + \lambda\lambda^{-1}(\eta'\eta^{-1} - 1)] \bmod (q^{nm}-1)$$

where k is an arbitrary positive integer and λ^{-1} and η^{-1} are integers satisfying $\lambda\lambda^{-1} = 1 \bmod N_y$ and $\eta\eta^{-1} = 1 \bmod p_y$, respectively.

A $\gamma\beta$-array and its symmetric images with respect to the x- and/or y-axes may generally be regarded to be the same, since all of them can easily be obtained if any one of them

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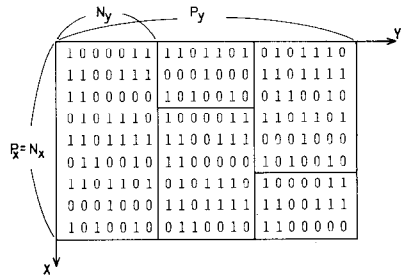


Fig. 5. Example of the macroscopic parameters of a $\gamma\beta$-array ($q = 2$, $n = 6$, $m = 1$, $\mu = 7$, $\lambda = 3$, $\eta = 2$, $N_x = p_x = 9$, $p_y = 21$, $N_y = 7$, $-\eta\lambda \bmod p_x = 3$).

is given. Concerning the symmetric images of a $\gamma\beta$-array, we have proved the following theorem.

Theorem 13: The two-dimensional code C_B that is the symmetric image of a $\gamma\beta$-array code C_A: $\langle n,m,\mu,\lambda,\eta,\alpha\rangle$ with respect to the x axis is a $\gamma\beta$-array code and is represented by C_B: $\langle n,m,\mu,\lambda,e_\alpha - \eta,\alpha\rangle$.

The two-dimensional code C_C that is the symmetric image of C_A with respect to the y axis is a $\gamma\beta$-array code and is represented by C_C: $\langle n, m, \mu, \lambda, e_\alpha - \eta, \alpha^{-1}\rangle$, where $e_\alpha = q^{nm} - 1$ is the order of α.

Corollary 3: The two-dimensional code C_D that is the symmetric image with respect to both x and y axes of the code C_A is a $\gamma\beta$-array code and is represented by C_D: $\langle n,m,\mu,\lambda,\eta,\alpha^{-1}\rangle$.

The Characteristic Arrays and the Coset Decomposition of the $\gamma\beta$-Array

Let $f(x,y)$ be a $\gamma\beta$-array whose period is (p_x,p_y). Then let q_x and q_y be integers satisfying

$$\gcd(p_x,q_x) = 1, \qquad 1 \le q_x < p_x \tag{45}$$

$$\gcd(p_y,q_y) = 1, \qquad 1 \le q_y < p_y. \tag{46}$$

Let us define a set Q to be the set of all ordered pairs of q_x and q_y, namely

$$Q = \{(q_x,q_y) \mid q_x \in Q_x,\ q_y \in Q_y\},$$

where $Q_x = \{q_x\}$ and $Q_y = \{q_y\}$ are the sets of all q_x and q_y satisfying (45) and (46), respectively. Obviously Q has $\phi(p_x)\phi(p_y)$ elements, where ϕ is Euler's function.

A transformation $\pi(q_x,q_y)$ of a $\gamma\beta$-array is defined by

$$\pi(q_x,q_y)f(x,y) = f(x^{q_x}, y^{q_y}) \tag{47}$$

and is called a decimation of a $\gamma\beta$-array. From (45) and (46) it is easily shown that a decimation is equivalent to a permutation of the coefficients of the $\gamma\beta$-array.

The following theorem forms the foundation of this section.

Theorem 14: $\pi(q_x,q_y)f(x,y)$ is a $\gamma\beta$-array whose parameters n, m, μ, and λ are the same as those of $f(x,y)$.

Next, the characteristic arrays of a $\gamma\beta$-array are studied.

Lemma 15: The macroscopic parameters of the $\gamma\beta$-array $f(x,y)$ and the $\gamma\beta$-array $\pi(q_x,q_y)f(x,y)$ are the same if and only if

$$q_x^{-1} = v \bmod p_x$$
$$q_y^{-1} = v \bmod p_y,$$

where v is some integer relatively prime to $q^{nm} - 1$, and q_x^{-1} and q_y^{-1} are the integers defined by $q_x^{-1}q_x = 1 \bmod p_x$ and $q_y^{-1}q_y = 1 \bmod p_y$, respectively. There are $\phi(q^{nm} - 1)$ decimations that do not alter the macroscopic parameters of the $\gamma\beta$-array.

Lemma 16: The $\gamma\beta$-array $\pi(q_x,q_y)f(x,y)$ is a cyclic shift of the $\gamma\beta$-array $f(x,y)$ if and only if

$$q_x^{-1} = q^k \bmod p_x$$
$$q_y^{-1} = q^k \bmod p_y,$$

where k is some positive integer. There are nm such decimations.

The $\gamma\beta$-array whose image with an arbitrary decimation of Lemma 16 is the original array itself is called a characteristic array.

Theorem 17: There are $q - 1$ characteristic arrays among all $q^{nm} - 1$ cyclic shifts of a $\gamma\beta$-array.

Theorem 18: The array that is obtained by a decimation of a characteristic array of a $\gamma\beta$-array is also a characteristic array of a $\gamma\beta$-array.

Hereafter, q_x and q_y are regarded as representing reduced residue classes modulo p_x and p_y, respectively. Then Q is a direct product of multiplicative groups $Q_x = \{q_x\}$ and $Q_y = \{q_y\}$, and is a commutative group with multiplication $(q_x,q_y) * (q_x',q_y') = (q_xq_x',q_yq_y')$. The transformation group $\{\pi(q_x,q_y)\}$ is isomorphic to Q when multiplication is defined as successive transformations.

Let C_0 be a set of the elements of Q that corresponds to a decimation that yields a cyclic shift of the $\gamma\beta$-array (cf. Lemma 16). Obviously C_0 is a subgroup of Q. Similarly, C_0' is defined to be a set of the elements of Q that corresponds to a decimation that does not alter the macroscopic parameters of the $\gamma\beta$-array (cf. Lemma 15). C_0' is a subgroup of Q. Furthermore, C_0 is a subgroup of C_0'.

Theorem 19: A $\gamma\beta$-array $\pi(q_x,q_y)f(x,y)$ is a cyclic shift of a $\gamma\beta$-array $\pi(q_x',q_y')f(x,y)$ if and only if (q_x,q_y) and (q_x',q_y') belong to the same residue class of a subgroup C_0 of Q, where $f(x,y)$ is a $\gamma\beta$-array. Further, the macroscopic parameters of $\pi(q_x,q_y)f(x,y)$ and $\pi(q_x',q_y')f(x,y)$ are the same if and only if (q_x,q_y) and (q_x',q_y') belong to the same residue class of a subgroup C_0' of Q.

Theorem 20: Let $Q' = \{(i,j) \mid 0 \le i < p_x,\ 0 \le j < p_y\}$. The element (i,j) is regarded as representing the (i,j)th position of a $\gamma\beta$-array. Let $Q'(\supset Q)$ be decomposed into residue classes (including improper cosets) of the subgroup C_0 of Q. Then the positions of a characteristic array of a $\gamma\beta$-array have the same coefficient (element of $GF(q)$) if they belong to the same residue class.

Theorem 21: If a $\gamma\beta$-array $f'(x,y)$ has the same parameters n, m, μ, and λ as a $\gamma\beta$-array $f(x,y)$, we can construct $f'(x,y)$ from $f(x,y)$ by applying a decimation and executing a cyclic shift.

Theorem 22: When parameters n, m, μ, and λ satisfying conditions (10), (11), and (12) are given, there are

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TABLE I
PARAMETERS OF THE $\gamma\beta$-ARRAYS OVER $GF(2)$ WITH AREAS N_xN_y NOT GREATER THAN 1023 AND NOT LESS THAN 3

n	m	μ	λ	$p_x=N_x$	p_y	N_y	p_y/N_y	
nm = 2								
1	2	1	1	1	3	3	1	M*
2	1	1	1	3	3	1	3	
2	1	1	3	3	1	1	1	M*
nm = 3								
1	3	1	1	1	7	7	1	M*
3	1	1	1	7	7	1	7	
3	1	1	7	7	1	1	1	M*
nm = 4								
1	4	1	1	1	15	15	1	M*
2	2	1	1	3	15	5	3	
2	2	1	3	3	5	5	1	M
4	1	1	1	15	15	1	15	
4	1	1	3	15	5	1	5	
4	1	1	5	15	3	1	3	
4	1	1	15	15	1	1	1	M*
4	1	3	1	5	15	3	5	
4	1	3	5	5	3	3	1	M
nm = 5								
1	5	1	1	1	31	31	1	M*
5	1	1	1	31	31	1	31	
5	1	1	31	31	1	1	1	M*
nm = 6								
1	6	1	1	1	63	63	1	M*
2	3	1	1	3	63	21	3	
3	2	1	1	7	63	9	7	
3	2	1	7	7	9	9	1	M
6	1	1	1	63	63	1	63	
6	1	1	3	63	21	1	21	
6	1	1	7	63	9	1	9	
6	1	1	9	63	7	1	7	
6	1	1	21	63	3	1	3	
6	1	1	63	63	1	1	1	M*
6	1	3	1	21	63	3	21	
6	1	3	7	21	9	3	3	
6	1	7	1	9	63	7	9	
6	1	7	3	9	21	7	3	
6	1	7	9	9	7	7	1	M
nm = 7								
1	7	1	1	1	127	127	1	M*
7	1	1	1	127	127	1	127	
7	1	1	127	127	1	1	1	M*
nm = 8								
1	8	1	1	1	255	255	1	M*
2	4	1	1	3	255	85	3	
2	4	1	3	3	85	85	1	M
4	2	1	1	15	255	17	15	
4	2	1	3	15	85	17	5	
4	2	1	5	15	51	17	3	
4	2	1	15	15	17	17	1	M
4	2	3	1	5	255	51	5	
4	2	3	5	5	51	51	1	M
8	1	1	1	255	255	1	255	
8	1	1	3	255	85	1	85	
8	1	1	5	255	51	1	51	
8	1	1	15	255	17	1	17	
8	1	1	17	255	15	1	15	
8	1	1	51	255	5	1	5	
8	1	1	85	255	3	1	3	
8	1	1	255	255	1	1	1	M*
8	1	3	1	85	255	3	85	
8	1	3	5	85	51	3	17	
8	1	3	17	85	15	3	5	
8	1	3	85	85	3	3	1	M
8	1	5	1	51	255	5	51	
8	1	5	3	51	85	5	17	
8	1	5	17	51	15	5	3	
8	1	5	51	51	5	5	1	M
8	1	15	1	17	255	15	17	
8	1	15	17	17	15	15	1	M
nm = 9								
1	9	1	1	1	511	511	1	M*
3	3	1	1	7	511	73	7	
3	3	1	7	7	73	73	1	M
9	1	1	1	511	511	1	511	
9	1	1	7	511	73	1	73	
9	1	1	73	511	7	1	7	
9	1	1	511	511	1	1	1	M*
9	1	7	1	73	511	7	73	
9	1	7	73	73	7	7	1	M

TABLE I (*Cont'd.*)
PARAMETERS OF THE $\gamma\beta$-ARRAYS OVER $GF(2)$ WITH AREAS N_xN_y NOT GREATER THAN 1023 AND NOT LESS THAN 3

n	m	μ	λ	$p_x=N_x$	p_y	N_y	p_y/N_y	
nm = 10								
1	10	1	1	1	1023	1023	1	M*
2	5	1	1	3	1023	341	3	
2	5	1	3	3	341	341	1	M
5	2	1	1	31	1023	33	31	
5	2	1	31	31	33	33	1	M
10	1	1	1	1023	1023	1	1023	
10	1	1	3	1023	341	1	341	
10	1	1	11	1023	93	1	93	
10	1	1	31	1023	33	1	33	
10	1	1	33	1023	31	1	31	
10	1	1	93	1023	11	1	11	
10	1	1	341	1023	3	1	3	
10	1	1	1023	1023	1	1	1	M*
10	1	3	1	341	1023	3	341	
10	1	3	11	341	93	3	31	
10	1	3	31	341	33	3	11	
10	1	3	341	341	3	3	1	M
10	1	11	1	93	1023	11	93	
10	1	11	3	93	341	11	31	
10	1	11	31	93	33	11	3	
10	1	11	93	93	11	11	1	M
10	1	31	1	33	1023	31	33	
10	1	31	3	33	341	31	11	
10	1	31	11	33	93	31	3	
10	1	31	33	33	31	31	1	M
10	1	93	1	11	1023	93	11	
10	1	93	11	11	93	93	1	M

An M in the last column indicates that the $\gamma\beta$-array is a $\gamma\beta$-M-array, and M^* indicates that a period of the $\gamma\beta$-array is an M-sequence.

$\phi[(q^n - 1)/\mu\lambda]$ $\gamma\beta$-array codes whose macroscopic structures are different. The number of $\gamma\beta$-array codes is

$$\frac{\phi(q^{nm} - 1)}{nm}\,\phi\left(\frac{q^n - 1}{\mu\lambda}\right)$$

and the number of $\gamma\beta$-arrays is

$$(q^{nm} - 1)\,\frac{\phi(q^{nm} - 1)}{nm}\,\phi\left(\frac{q^n - 1}{\mu\lambda}\right).$$

Remark: It can be shown that

$$\phi(q^{nm} - 1)\phi\left(\frac{q^n - 1}{\mu\lambda}\right) = \phi\left(\frac{q^n - 1}{\mu}\right)\phi\left(\frac{q^{nm} - 1}{\lambda}\right) = \phi(p_x)\phi(p_y).$$

VI. Conclusions

We have given a method of construction and many interesting characteristics of two-dimensional linear recurring arrays having maximum-area matrices. Most of the characteristics of the $\gamma\beta$-arrays we have described can be regarded as two-dimensional analogs of those of the usual M-sequences. The characteristic peculiar to $\gamma\beta$-arrays that is not present in M-sequences is the difference between the two concepts of period and area of the maximum-area matrices.

A problem left unsolved is whether there exist arrays having maximum-area matrices other than $\gamma\beta$-arrays. No such example has been found so far, though it has not been proved that $\gamma\beta$-arrays exhaust all of the linear recurring arrays having maximum-area matrices.

Acknowledgment

The authors wish to thank Prof. Y. Taki, Prof. Y. Yasuda, and Dr. K. Ito of the University of Tokyo for their very stimulating discussions and encouragement during the course of this study.

Appendix

The parameters of the $\gamma\beta$-arrays over $GF(2)$ with area N_xN_y not greater than 32 767 and not less than 3 have been calculated

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by digital computer. The $\gamma\beta$-arrays with area not greater than 1023 are tabulated in Table I.

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On the Number of Information Symbols in Polynomial Codes

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***Abstract*—Polynomial codes and their dual codes as introduced by Kasami, Lin, and Peterson have considerable algebraic and geometric structure. It has been shown that these codes contain many well-known classes of cyclic codes as subclasses, such as BCH codes, projective geometry codes (PG codes), Euclidean geometry codes (EG codes), and generalized Reed–Muller codes (GRM codes). In this paper, combinatorial expressions for the number of information symbols and parity-check symbols in polynomial codes are derived. The results are applied to two important subclasses of codes, the PG codes and EG codes.**

I. Introduction

POLYNOMIAL codes and their dual codes as introduced by Kasami *et al.* [1] have considerable inherent mathematical structure [1]–[5]. It has been shown [1], [2] that these codes contain several important classes of cyclic codes as subclasses, such as BCH codes [6], [7], finite projective geometry codes (PG codes) [8]–[12], Euclidean geometry codes (EG codes) [8], [11]–[13], and generalized Reed–Muller codes (GRM codes) [14]. In this paper, the problem of enumerating the number of information symbols in polynomial codes and their dual codes is studied. First, combinatorial expressions for the number of information symbols and the number of parity-check symbols in polynomial codes are derived. Next, we apply these results to projective geometry codes and Euclidean geometry codes.

In order to present our results, it is necessary to give a brief review of polynomial codes and their duals. For details, the reader is referred to [1]–[4].

Let m and s be two positive integers. Let $GF(q^{ms})$ be the extension field of the field $GF(q^s)$, where q is a power of a prime p. Let α be a primitive element of $GF(q^{ms})$. Any nonzero element α^j in $GF(q^{ms})$ can be expressed as

$$\alpha^j = a_{1j} + a_{2j}\alpha^1 + a_{3j}\alpha^2 + \cdots + a_{mj}\alpha^{m-1} \tag{1}$$

for $0 \le j < q^{ms} - 1$, where a_{ij} is in $GF(q^s)$. There is a one-to-one correspondence between α^j and the m-tuple $\boldsymbol{A}_j = (a_{1j},a_{2j},\cdots,a_{mj})$. The m-tuple $\boldsymbol{A}_j$ is called the *coordinate vector* of α^j. Let b be a factor of $q^s - 1$. We define two positive integers, z and n, as follows:

$$z = (q^s - 1)/b \tag{2}$$

$$n = (q^{ms} - 1)/b. \tag{3}$$

For the purposes of clarity, the definitions of b, z, and n will be recalled frequently throughout this paper.

Let $\boldsymbol{X} = (X_1,X_2,\cdots,X_m)$, where X_i is a variable over $GF(q^s)$. Let μ be a positive integer less than mz. Define $Q_m(\mu,b)$ as the set of polynomials in m variables,

$$f(\boldsymbol{X}) = \sum_{v_1}\sum_{v_2}\cdots\sum_{v_m} C_{v_1v_2\cdots v_m} X_1^{v_1} X_2^{v_2}\cdots X_m^{v_m} \tag{4}$$

for which

1) $C_{v_1v_2\cdots v_m} \in GF(q^s)$

2) $0 \le v_i < q^s, \quad 1 \le i \le m$

3) $\sum_{i=1}^{m} v_i = lb, \quad 0 \le l \le \mu$

4) $f(a_{1j},a_{2j},\cdots,a_{mj}) \in GF(q), \quad 0 \le j < n,$

where $(a_{1j},a_{2j},\cdots,a_{mj})$ is the coordinate vector of α^j. For each polynomial $f(\boldsymbol{X})$ in $Q_m(\mu,b)$, a vector $\boldsymbol{v}(f)$ is defined as follows:

$$\boldsymbol{v}(f) = (v_0,v_1,v_2,\cdots,v_{n-1}), \tag{5}$$

Manuscript received November 1, 1971; revised April 10, 1972. This work was supported by NSF Grant GK-25128.

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